

Digest: Hamiltonian for the zeros of the Riemann zeta function

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1 Introduction

This document digests the paper addressing the Hilbert-Pólya conjecture. The authors construct an explicit quantum mechanical Hamiltonian whose eigenvalues mirror the imaginary parts of the nontrivial zeros of the Riemann zeta function.

2 The Hamiltonian

The proposed Hamiltonian operator $\hat{H}$ is given by:

$$\hat{H} = \frac{1}{1 - e^{-i\hat{p}}}(\hat{x}\hat{p} + \hat{p}\hat{x})(1 - e^{-i\hat{p}}) \quad (1)$$

This Hamiltonian is a similarity transformation of the Berry-Keating Hamiltonian $\hat{h}_{\text{BK}} = \hat{x}\hat{p} + \hat{p}\hat{x}$ via the nonlocal difference operator $\Delta = 1 - e^{-i\hat{p}}$.

3 $\mathcal{PT}$ Symmetry and the Riemann Hypothesis

Although $\hat{H}$ is not Hermitian in the conventional sense, the operator $i\hat{H}$ is $\mathcal{PT}$ symmetric (parity-time reflection invariant under momentum exchange). If $i\hat{H}$ has maximally broken $\mathcal{PT}$ symmetry, its eigenvalues are pure-imaginary complex-conjugate pairs. Consequently, the eigenvalues of $\hat{H}$ would be strictly real, thereby proving the Riemann Hypothesis.

4 Eigenfunctions

The eigenfunctions of $\hat{H}$, subjected to the boundary condition $\psi_n(0) = 0$, evaluate to the Hurwitz zeta function $\psi_z(x) = -\zeta(z, x+1)$. The boundary condition forces z to correspond precisely to the nontrivial zeros of the Riemann zeta function.